

IoT-Based Laser Security System: Real-Time Intrusion Alerting via ESP32 and Telegram Bot API

1. Abstract

This project presents the development of a low-cost IoT-based laser security system designed for real-time intrusion detection and remote alerting. The system uses an ESP32 microcontroller, a laser diode, and an LDR sensor to form an optical tripwire that continuously monitors a protected area. When the laser beam is interrupted, the ESP32 detects the change in light intensity and immediately activates a buzzer alarm while simultaneously sending a real-time notification to the user through Telegram Messenger using the Telegram Bot API over Wi-Fi. The system eliminates the need for expensive GSM modules and provides a simple, reliable, and scalable security solution that allows users to monitor locations remotely from anywhere with internet access. Due to its low cost and ease of implementation, the system can be effectively used in homes, offices, laboratories, and other restricted areas for improved security monitoring.

2. Introduction

In today's world, ensuring the safety of homes, offices, and restricted areas has become increasingly important. Traditional security systems often involve expensive equipment, complex wiring, or subscription-based monitoring services, making them less accessible to many users. With the rapid advancement of the Internet of Things (IoT), it has become possible to develop smart and affordable security systems that offer real-time monitoring and instant alerts.

This project presents an IoT-Based Laser Security System that detects unauthorized entry using an optical tripwire mechanism formed by a laser diode and an LDR sensor. An ESP32 microcontroller continuously monitors the light intensity, and when the laser beam is interrupted, it activates a buzzer alarm and sends an instant notification to the user through Telegram Messenger via Wi-Fi. The system provides a simple, reliable, and cost-effective solution for remote security monitoring.

3. Problem Statement

Many conventional security systems rely on expensive hardware, complex installations, or paid monitoring services, making them difficult for small homes and organizations to adopt. In addition, several low-cost systems lack real-time remote alerts. Therefore, there is a need for an affordable, reliable, and easy-to-implement security solution that can detect intrusions instantly and notify users remotely through internet-based communication platforms.

4. Objectives of the Project

- To develop a low-cost IoT laser security system using a laser and LDR sensor to detect intrusions.
- To detect beam interruption using the ESP32 microcontroller.
- To send instant alerts via buzzer and Telegram notifications.
- To enable remote monitoring through Wi-Fi and Telegram Bot API.
- To provide a simple, reliable, and scalable security solution for various environments.

5. Literature Review

Recent studies have shown that IoT-based security systems can provide efficient real-time monitoring and remote alerts. The paper “Laser Light Security Alert System using ESP32-Cam with Integration of Telegram IoT” (IJARSCT, 2025) demonstrates how an ESP32-based system can detect intrusions and send notifications through Telegram. Similarly, “Security System in a Private Room Using ESP32 Microcontroller with Telegram Bot” (ResearchGate, 2025) highlights the effectiveness of Telegram bots for remote monitoring and instant alert communication.

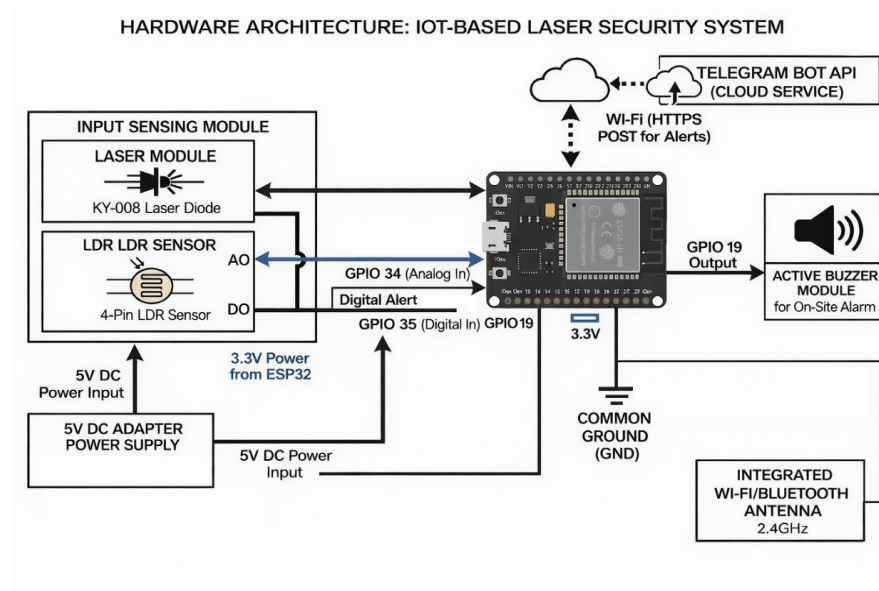
Another study titled “Laser Based Security System” (IARJSET, 2021) explains the use of a laser beam and light sensor to create an optical tripwire for intrusion detection. Practical implementations of similar concepts are demonstrated in online resources such as ESP32 Laser Tripwire Alarm – Robocraze and DIY ESP32 Telegram Security Bot – Afghan Robotic Team which show how ESP32 sensors and Telegram bots can be integrated to send real-time alerts for security applications.

6. Proposed Methodology

The proposed system uses an ESP32 microcontroller, a laser diode, an LDR sensor, and an active buzzer to detect intrusions. A laser beam is directed toward the LDR to form an optical tripwire across the protected area. The ESP32 continuously monitors the light intensity using its analog input. When the beam is interrupted, the sudden drop in light intensity is detected as an intrusion.

Once detected, the ESP32 activates a buzzer alarm and simultaneously sends a real-time notification to the user via Telegram Messenger using the Telegram Bot API over Wi-Fi. This approach provides both local alerts and remote monitoring, making the system suitable for basic security applications.

6.1 Block Diagram



6.2 Hardware Components

- 1) ESP32 microcontroller
- 2) Buzzer
- 3) KY-008 Laser Diode Module
- 4) LDR Sensor Module
- 5) Power Supply Unit (5V Adapter)

6.3 Software Components

- 1) Arduino IDE: Used for programming the ESP32

- 2) UniversalTelegramBot Library: Facilitates the HTTPS communication with the Telegram API.
- 3) ArduinoJson Library: Used for parsing complex message data received from the cloud.
- 4) Telegram Messenger Application – Installed on the user’s smartphone to receive intrusion alerts.

7. Expected Outcomes

The proposed system is expected to successfully detect unauthorized entry by monitoring the laser beam using an LDR sensor. Whenever the beam is interrupted, the ESP32 will immediately trigger a buzzer alarm and send a notification to the user through Telegram. This project aims to provide a simple, affordable, and reliable security solution that allows users to monitor their premises remotely in real time.

8. Applications

- Home Security System
- Office and Workplace Security
- Laboratory and Research Facility Protection
- Warehouse and Storage Monitoring
- Bank Locker and Restricted Area Security

References

- [1] “[Laser Light Security Alert System using ESP32-Cam with Integration of Telegram IoT](#)” (IJARSCT, 2025)
- [2] “[Security System in a Private Room Using ESP32 Microcontroller with Telegram Bot](#)” (ResearchGate, 2025)
- [3] “[Laser Based Security System](#)” (IARJSET, 2021)
- [4] “[ESP32 Laser Tripwire Alarm](#)” by Robocraze.
- [5] “[DIY ESP32 Telegram Security Bot](#)” by the Afghan Robotic Team.